

Hints & Solutions

16. If an electron attempts to come out of the metal, the metal surface acquires a positive charge and pulls the electron back to the metal. Consequently, the electron can come out of the metal surface only if it has got sufficient energy to overcome the attractive pull.
17. In the discharge tube a fluorescent glow is seen on the glass opposite to cathode. The colour of glow of the glass depended on the type of glass. It being yellowish-green for soda glass.
18. X-rays does not have charge, they are simple electromagnetic radiation, hence, it cannot be deflected by applying electric and magnetic field.
19. The value of e/m was found to be independent of the nature of the material or metal used as the cathode (emitter) or the gas introduced in the discharge tube.
20. Hallwachs and Lenard also observed that when ultraviolet light fell on the emitter plate, no electrons were emitted at all when the frequency of the incident light was smaller than a certain minimum value, called the threshold frequency. This minimum frequency depends on the nature of the material of the emitter plate.
21. Some alkali metals such as, lithium, sodium, potassium, caesium and rubidium are sensitive even to visible light. All these photosensitive substances emit electrons when they are illuminated by light.
22. Positive charge on a positively charged zinc plate was found to be further enhanced when it was illuminated by ultraviolet light. So, leaves of electroscope will move further apart.
23. Stopping potential is more negative for higher frequency.
24. Photons are electrically neutral and so are not deflected by strong electric or magnetic field.
25. This wavelength is so small that it is beyond any measurement. This is reason why macroscopic objects in our daily life do not show wavelike properties.
26. The wave properties of electron have been utilised in the design of electron microscope which is a great improvement, with higher resolution, over the optical microscope.

27. Momentum of a photon of energy $h\nu$

$$p = \frac{h\nu}{c} = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{p} \quad \dots(i)$$

Also, by de-Broglie wavelength formula

$$\lambda = \frac{h}{p} \quad \dots(ii)$$

Both agrees to same value.

28. Cathode rays are streams of negatively charged particles.
29. The particles speed ranges from $0.1 C$ to $0.2 C$. Here C is the speed of light in vacuum which is 3×10^8 m/s.
30. Experimentally it is found that, if frequency of the incident radiation exceeds the threshold frequency, the photoelectric emission starts instantaneously without any apparent time lag, even if the incident radiation is very dim. It is now known that emission starts in a time of the order of 10^{-9} sec or less.
31. $eV_0 = h\nu - \phi$
33. Photo electric effect is instantaneous process without any apparent time ($\sim 10^{-9}$ s or less) even when incident light is made dim.
34. Photo cell is used to detect light.
35. Work function is maximum for platinum & minimum for cesium.
36. Photon may be absorbed or a new photon may be created.
37. For caesium $\phi = 2.14$ eV
Energy of ultraviolet & visible light is greater than work function of caesium so they can cause photo emission.
38. G.P. Thomson observed diffraction effects with beams of electrons scattered by crystals.
39. (a) In $\lambda = \frac{h}{p}$ equation $\rightarrow$
 λ is characteristic of wave & p (momentum) is

characteristic of particle.

(b) Mass of macroscopic particles is high so de-Broglie wavelength is very small.

(c) $\lambda = \frac{h}{p}$ is applicable for material particle &

photons as well.

40. Intensity of electron beam is proportional to number of e^- per unit time entering detector so current is proportional to intensity.

41. Photons have momentum given by $\frac{h}{\lambda}$

It is not material particle & it is packet of energy. Photon energy depends on frequency of light.

42. Wave nature of e^- is utilized in e^- microscope.

43. Minimum frequency for photo electric effect is threshold frequency ν_0 which depends on work function of metal by equation

$$\phi = h\nu_0$$

44. For a given source,

Intensity $\propto$ number of photons per sec.

$\propto$ number of photo e^- per sec.

$\propto$ current

46.
$$\frac{e}{m} = \frac{1.6 \times 10^{-19}}{9.1 \times 10^{-31}} = 1.76 \times 10^{11} \text{ C / kg}$$